Global Spatial and Local Spectral Similarity-Based Manifold Learning Group Sparse Representation for Hyperspectral Imagery Classification

Haoyang Yu, *Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Wenzhi Liao, *Senior Member, IEEE*, Bing Zhang, *Senior Member, IEEE*, Lina Zhuang, *Student Member, IEEE*, Meiping Song, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Spectral–spatial framework has been widely applied for hyperspectral image classification task. Some well-established models, such as group sparse representation (GSR), have gained a certain advance but still mainly focus on the usage of local spatial similarity and neglect the nonlocal spatial information. Recently, nonlocal self-similarity (NLSS) has been exploited to support the spatial coherence tasks. However, current NLSS-based methods are biased toward the direct use of nonlocal spatial information as a whole, while the underlying spectral information is not well exploited. In this article, we proposed a novel method to exploit local spectral similarity through nonlocal spatial similarity, with the integration of local spatial consistency in a single framework. Specifically, the proposed approach first exploits the NLSS by searching the nonoverlapped similar patches in defined scopes. Then, spectral similarity is determined locally within the found patches. After that, the found similar data and the original data are fused in a designed pattern. Finally, the GSR-based classifier (GSRC) is applied to process the fused data characterized by the manifold learning algorithm. The experimental results based on three real hyperspectral data sets demonstrate the efficiency of the proposed method, with improvements over the other related nonlocal or local similarity-based methods.**

***Index Terms*— Classification, group sparse representation (GSR), hyperspectral image, nonlocal self-similarity (NLSS).**

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H. Yu and B. Zhang are with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: yuhy@dlmu.edu.cn; zb@radi.ac.cn).

L. Gao is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: gaolr@radi.ac.cn).

W. Liao is with the Department of Electronic and Electrical Engineering, University of Strathclyde, Glasgow G1 1XW, U.K. (e-mail: wenzhi.liao@strath.ac.uk).

L. Zhuang is with the Department of Mathematics, Hong Kong Baptist University, Hong Kong (e-mail: linazhuang@hkbu.edu.hk).

M. Song is with the Center of Hyperspectral Imaging in Remote Sensing (CHIRS), Information Science and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: smping@163.com).

J. Chanussot is with the GIPSA-Lab, CNRS, Grenoble INP, Université Grenoble Alpes, F-38000 Grenoble, France, and also with the Faculty of Electrical and Computer Engineering, University of Iceland, 101 Reykjavik, Iceland (e-mail: jocelyn.chanussot@gipsa-lab.grenoble-inp.fr).

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I. Introduction

HYPERSPECTRAL remote sensing is characterized by its high spectral resolution, which allows capturing both the image and the continuous spectrum [1]. This property brings a hyperspectral image with abundant information, which benefits obtaining more details of the characteristics of materials in a wide range of applications, especially for classification [2]–[5]. The purpose of classification is to apply certain classifiers to assign a unique label to each pixel vector such that it can be identified and defined by a given class [6]. By treating the spectrum of each pixel as the feature of a signal, several classic models have been proved to provide significant performances for hyperspectral image classification tasks [7], [8]. However, from a geographical point of view, the hyperspectral image is not just an unordered combination of signals. It actually contains underlying information about the properties of ground objects and the correlations between them, which can be exploited for more accurate discrimination [9]–[12].

In recent years, sparse representation (SR) has been widely exploited for classification tasks based on its appropriate use of features and representation of potential relationships among the pixels [13]. The basic assumption of SR-based classifiers (SRCs) is that the natural images can be sparsely coded by structural primitives [14]. For hyperspectral image, although the original scene is not sparse, it has been demonstrated that there are high correlations in both the spectral and spatial domains [15]–[17]. Therefore, the hyperspectral data can be represented sparsely [18]. SRC codes a testing pixel as a linear combination of the training samples in the dictionary and then classifies the testing pixel by evaluating which class leads to the minimum total residual error [19]. Several SR-based models have proved that it can provide rather plausible results compared with those of traditional classifiers [20]–[22]. However, SRC mainly exploits information in the spectral domain, and it does not take advantage of the spatial correlations in hyperspectral images that would allow for more accurate classification.

In order to incorporate the spatial information, the second generation of SRCs has appeared in the literature by integrating additional structure sparisty priors [23], [24]. These priors are based on exploiting the local spatial dependencies of pixels in neighborhood, the inherent structure property of the

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dictionary, or both [25], [26]. Among these, group SR-based classifier (GSRC) has been considered as a relatively complete model due to its joint representation of pixels in neighborhood and activation of group atoms instead of the individual ones in the dictionary and sparse coefficients. It circumvents the instability of sparse coefficients and achieves quite persuasive classification results for hyperspectral image [27], [28]. In comparison to the original counterparts, GSRC considers the spatial consistency of a group of pixels surrounding a target pixel [29]. However, it neglects the spatial correlation of other pixels outside the local area (i.e., nonlocal pixels), which can be further exploited for more comprehensive discrimination.

Nonlocal self-similarity (NLSS) has been widely used in natural image processing and holds the state of the art in image denoising [30], [31]. NLSS is motivated by a fact that natural images are self-similar, meaning that they contain many similar patches at different locations and scales [32], [33]. By exploiting the spatial dependence in a nonlocal area, it can offer references of global structure prior [34], [35]. Recently, several NLSS-based approaches have been designed for hyperspectral imagery classification [36], [37]. In [36], a nonlocal means (NL-means) algorithm based on NLSS was introduced into the maximum *a posteriori* (MAP) estimation and proven to be effective for hyperspectral classification. In [38], an extended Markov random field model is designed to combine multiple features with local and nonlocal spatial constraints. In [39], the NLSS was integrated with the joint sparsity model, where the similarity of nonlocal patches centered on the pixels in the neighborhood is computed to adjust the coefficients during the representation. Although NLSS-based methods have shown their properties for hyperspectral image processing, they are essentially biased toward the direct use of nonlocal spatial similar information as a whole, while the current fusion pattern remains instability and the inherent spectral information are not well exploited, which allows for more accurate classification.

In this article, we proposed a novel method to exploit local spectral similarity through nonlocal spatial similarity, with the integration of local spatial information in a single framework. Specifically, the proposed method can be implemented in four main steps: 1) a nonlocal search for spatially similar patches in defined scopes, finding nonoverlapped patches similar to the target patch containing the testing pixel and its surrounding pixels; 2) a local search for spectra within the found nonlocal patches, finding the most spectrally similar pixels to the one under investigation; 3) a fusion of global information, where the found nonlocal similar pixels and the pixels in the local neighborhood are fused in the designed pattern; and 4) a global structure-based representation, where the GSRC is applied to process the fused data characterized by the manifold learning algorithm. The proposed method, named global spatial and local spectral similarity (GSLS)-based manifold learning GSRC (GSLS-^{ML}GSRC), includes the contributions as follows.

1) The local spectral similarity is further exploited through nonlocal spatial similarity, which contributes to more accurate extraction of information for better discrimination.
2) The designed fusion pattern combines nonlocal and local spatial information, which provides global structure priors for more reliable response in the representation.
3) The NLSS is measured in defined scopes, and manifold learning algorithm is integrated for characterization, with consideration of both the practicability and the efficiency.

In the remainder of this article, Section II formulates the related models. Section III presents the proposed method. Section IV evaluates the classification performance. Section V concludes this article.

II. Related Models Description

A. Sparse Representation-Based Models

Although the original hyperspectral data are not sparse, it has been demonstrated that there are high correlations in the adjacent bands in the spectral domain and in neighboring pixels in the spatial domain [40], [41]. Therefore, the underlying hyperspectral image is essentially low rank and can be represented sparsely. SRCs represent a testing pixel via a linear combination of the available labeled samples, also called a dictionary, such that to determine the class by evaluating the errors between the obtained approximations and the one under investigation. With the assumption that neighboring pixels tend to belong to the same class, joint SRC (JSRC) is proposed to incorporate the local spatial information for a higher level constraint [42]. However, obstacles remain, such as the instability of sparse coefficients. Under this case, GRSC is designed to further exploit the inherent group-structured property of the dictionary coefficients [26], [43].

1) Sparse Representation-Based Classification: Given a hyperspectral image denoted as $\mathbf{X}$ with a total of N pixels, B spectral bands, and containing K thematic classes, $\mathbf{x}_{i,j}$ represents a testing pixel with coordinate of (i, j). For $\mathbf{x}_{i.j}$, the SRC model can be expressed as follows:

$$\arg\min_{\boldsymbol{\alpha}} \|\mathbf{x}_{i,j} - \mathbf{D}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_1 \tag{1}$$

where $\mathbf{D}$ is defined as a dictionary composed of labeled data, $\boldsymbol{\alpha}$ is a weight vector, and λ is a regularization parameter adjusting the sparisty of $\boldsymbol{\alpha}$. The main idea of SRC is to estimate the weight vector $\boldsymbol{\alpha}$ such that $\|\mathbf{x}_{i,j} - \mathbf{D}\boldsymbol{\alpha}\|_2^2$ is minimized with its l_0-norm $\|\boldsymbol{\alpha}\|$ being minimized. Here, the l_0-norm calculates the number of nonzero elements in a vector, of which the optimization is an NP-hard problem. Therefore, it is usually replaced with a l_1-nom regularization, which performs the sum of the absolute values in a vector, and can be solved by the orthogonal matching pursuit (OMP) [44]. After obtaining $\boldsymbol{\alpha}$, the class label of $\mathbf{x}_{i,j}$ is determined according to the class that minimized the class-specific residual under l_2-norm constraint

$$\mathrm{class}(\mathbf{x}_{i,j}) = \arg\min_{k} \|\mathbf{x}_{i,j} - \mathbf{D}\delta_k(\boldsymbol{\alpha})\|_2^2 \tag{2}$$

where $\delta_k(\boldsymbol{\alpha})$ is an indicator operator, whose value is zero for the atoms in $\boldsymbol{\alpha}$ that does not belong to class k ($k \in [1, K]$).

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2) Joint SRC: SRC mainly focuses on the correlated spectral characteristics in the spectral domain. However, it does not make use of the potential relationships among the pixels in the spatial domain. With the assumption that neighboring pixels usually consist of similar materials and tend to belong to the same class, the spatial correlation in the neighborhood can be directly incorporated through a JSRC model for improvement. Let $\mathbf{X}_{i,j} \in \mathbb{R}^{B\times t}$ represent a matrix whose columns correspond to t pixels in a spatial neighborhood with a centered pixel of $\mathbf{x}_{i,j}$. JSRC assumes that the underlying sparse vectors associated with $\mathbf{X}_{i,j}$ share a common support, such that to strengthen the sparse constraint of representation. Therefore, the objective function of JSRC can be expressed as

$$\arg\min_{\mathbf{A}} \|\mathbf{X}_{i,j} - \mathbf{D}\mathbf{A}\|_F^2 + \lambda\|\mathbf{A}\|_{1,2} \tag{3}$$

where $\mathbf{A}$ is a weight matrix whose columns are t weight vectors correspond to pixels in $\mathbf{X}_{i,j}$. Here, $\|\mathbf{A}\|_{1,2} = \sum_{i=1}^{n} \mathbf{a}^i$ is a $l_{1,2}$-norm regularization, where $\mathbf{a}^i$ represents the ith row of $\mathbf{A}$ and n is the number of labeled samples in $\mathbf{D}$, and can be solved by the simultaneously OMP (SOMP). After obtaining $\mathbf{A}$, the class label of $\mathbf{x}_{i,j}$ is determined by the minimum total residual error under the Frobenius-norm constraint

$$\text{class}(\mathbf{x}_{i,j}) = \arg\min_{k} \|\mathbf{X}_{i,j} - \mathbf{D}\delta_k(\mathbf{A})\|_F^2. \tag{4}$$

3) Group SRC: In SRC, the testing pixel is classified by evaluating the class-specific residual errors, which is calculated by a certain group of atoms in weight vector and dictionary. Inspired by this property, the GRSC model is designed to exploit the inherent structure-based priors of coefficients and dictionary and incorporated with the spatial dependences of pixels in neighborhood. The main idea of GSRC is to encourage sparse coefficients of only certain groups to be active and the remaining groups inactive so that the pixels are represented by a group of atoms instead of the individual ones. In GSRC, if we group atoms belonging to the same class, the matrix representing dictionary may be rearranged as $\mathbf{D} = (\mathbf{D}_1, \ldots, \mathbf{D}_K)$, where $\mathbf{D}_K$ denotes a subdictionary with each column vector representing a labeled sample from class K. Therefore, the objective function of GSRC can be expressed as

$$\arg\min_{\mathbf{A}} \|\mathbf{X}_{i,j} - \mathbf{D}\mathbf{A}\|_F^2 + \lambda \sum_{g\in G} \omega_g \|\mathbf{A}_g\|_2 \tag{5}$$

where $g \in \{G_1, \ldots, G_K\}$ indexes the K groups corresponding to the subdictionaries in $\mathbf{D}$, $\mathbf{A}_g$ represents the weight matrix of each group, and ω_g defines a compensation parameter for different sizes of groups. Here, $\sum_{g\in G} \|\mathbf{A}_g\|_2$ defines a group-level $l_{1,2}$-norm regularization and can be optimized by the Group Lasso algorithm [45]. After obtaining the coefficient matrix $\mathbf{A}$, the class label of $\mathbf{x}_{i,j}$ can be determined similar to (4). It should be noted from (3) and (5) that GSRC would degenerate to a JSRC model if the group size reduces to one.

B. Manifold Learning Algorithm

In the SR-based framework, the representation is actually performed under a feature-level response process. Therefore, an appropriate characterization is necessary and effective for the SR-based classification, which has been demonstrated in the literature [46], [47].

Manifold learning algorithms are based on the assumption that the input high-dimensional data can lie on a manifold subspace with a lower dimensionality [48]. According to the property of optimization algorithms for the eigenfunction, they can be divided into the linear models and the nonlinear ones. Although the nonlinear models provide impressive results on some data sets, their nonlinear property brings expensive computational cost [49]. Meanwhile, the linear models, such as neighborhood preserving embedding (NPE), have shown their advantages in preserving the local manifold structure on which the data may possibly reside and conducted in relatively low computational cost for the practical applications [50], [51]. For a hyperspectral image, manifold learning algorithms contribute to concise extraction and characterization of features, which circumvent the issue of dimensionality curse. Unlike some traditional dimension reduction methods, such as principal component analysis (PCA), which are bias toward preserving the global Euclidean structure, NPE represents each data point as a linear combination of neighboring pixels such that the local structure can be remained in the subspace [52], [53]. Specifically, NPE is implemented in three main steps.

1) Construction of adjacency graph, where the pixel $\mathbf{x}_{i,j}$ (following denoted as $\mathbf{x}_e$ with assumption that it is the eth pixel in $\mathbf{X}$) is connected to its M nearest neighbors by directed edges such that to construct an adjacency graph.
2) Determination of weights, which computes the weights on edges by minimizing the following objective functions:

$$\min_{H} \sum_{e} \left\| \mathbf{x}_e - \sum_{o} H_{eo}\mathbf{x}_o \right\|^2 \quad \text{s.t.} \sum_{o} H_{eo} = 1, \quad o \in [1, N] \tag{6}$$

where $\mathbf{H}$ represents a weight matrix, with H_{eo} defines the weight of edge between $\mathbf{x}_e$ and $\mathbf{x}_o$.
3) Manifold learning projection, which brings the projection by solving the following generalized eigenvector problem:

$$\mathbf{X}\mathbf{E}\mathbf{X}^T\mathbf{z} = \gamma\,\mathbf{X}\mathbf{X}^T\mathbf{z} \tag{7}$$

where $\mathbf{E} = (\mathbf{I} - \mathbf{H})^T(\mathbf{I} - \mathbf{H})$ is a symmetric and semipositive matrix and $\mathbf{I}$ is an identity matrix. Let $\mathbf{Z} = \mathbf{z}_1, \ldots, \mathbf{z}_d$ denote the solutions of (8) with increased order of the corresponding eigenvalues, and the projected image $\mathbf{X}^M = \mathbf{Z}^T\mathbf{X}$ can be obtained, where $\mathbf{X}^M = [\mathbf{x}_p^M|_{p=1}^N]$, and $\mathbf{x}_p^M$, i.e., $\mathbf{x}_{i,j}^M$, is a d-dimensional pixel vector ($d \ll B$).

III. Proposed Method

In comparison to SRC, GSRC incorporates structure priors with spatial dependences in the local area. In order to satisfy the homogeneity assumption, GSRC usually considers a small neighborhood around the testing pixel. If the neighborhood

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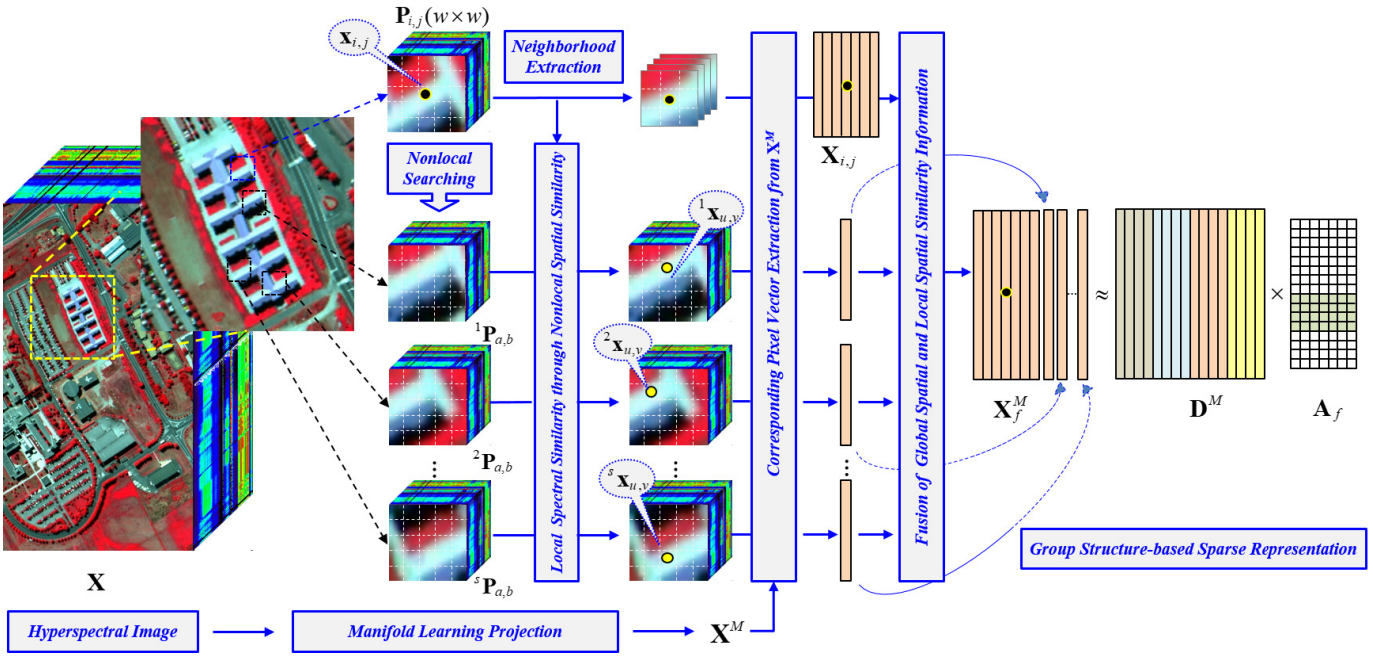


Fig. 1. General framework for the proposed GSLS-^{ML}GSRC.

locates at a complex scenario, such as the boundary of classes, the current spatial correlation will not be sufficient to provide an appropriate constraint of spatial consistency, which may lead to an error of response in the representation. Therefore, under this case, the nonlocal spatial information is particularly important, due to its supplement of spatial coherence from a nonlocal distribution. In our framework, we first consider that a larger patch contains the current neighborhood and find its nonlocal spatial similar patches in defined scopes. Then, we further locate the most spectrally similar pixels to the one under investigation in the found patches. After that, a fusion pattern is designed to incorporate the found nonlocal similar pixels and the pixels in the local neighborhood. Finally, the group SR (GSR)-based architecture is applied to process the fused data characterized by the manifold learning algorithm.

A. Local Spectral Similarity Through Nonlocal Spatial Similarity

As shown in Fig. 1, let $\mathbf{P}_{i,j}$ define a $w\times w$ pixel-sized patch with the center pixel located at $\mathbf{x}_{i,j}$. In order to satisfy the searching purpose, $\mathbf{X}$ is expanded to a $(r+w-1)\times(c+w-1)$ pixel-sized image, where r and c are the length of the original rows and columns. Then, we collect $w\times w$ pixel-sized patches centered on the pixels within a $v\times v$ pixel-sized search window around $\mathbf{x}_{i,j}$ ($v\le\min(r,c)$). Here, the definition of v is mainly for the balance between the efficiency and computation time. Besides, the collection process follows a principle that the nonlocal patches should be nonoverlapped with $\mathbf{P}_{i,j}$, thereby to guarantee that the contribution of the found information is apart from the ones in the local area. In order to satisfy this principle, we design a simple way to avoid the tedious process of judgment. As shown in Fig. 2, we apply a morphological operator of dilation to generate two masks centered on the location of $\mathbf{x}_{i,j}$ with the size of $(2w-1)\times(2w-1)$ and $v\times v$, respectively. The areas between two masks (colored in white in Fig. 2) are the locations, on which the patches centered satisfy the nonoverlapped principle in the defined region.

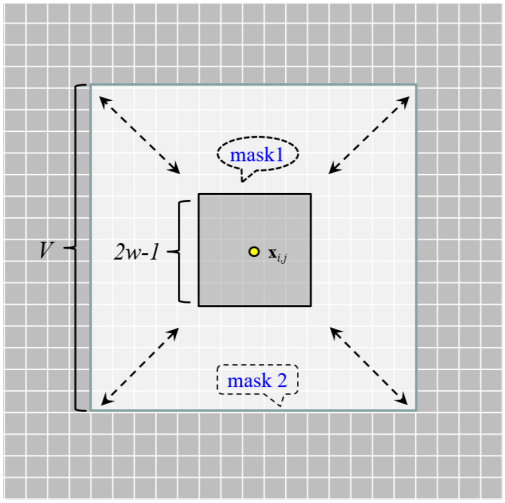


Fig. 2. Schematic of searching window.

Let $\mathbf{P}$ denote the set of collected patches, and the next process is to find the subset ${}^S\mathbf{P}=\{{}^s\mathbf{P}_{a,b}|s\in[1,S]\}$ from $\mathbf{P}$, where s indexes the similar patches. The similarity between $\mathbf{P}_{i,j}$ and $\mathbf{P}$ is measured by calculating the sum of Euclidian distances between the pixels in the corresponding locations as follows:

$$\operatorname*{dist}_{\mathbf{x}_p\in\mathbf{P}_{i,j},\mathbf{x}'_p\in{}^s\mathbf{P}_{a,b}}(\mathbf{P}_{i,j},{}^s\mathbf{P}_{a,b})=\sum_{p=1}^{w^2}\left\|\mathbf{x}_p,\mathbf{x}'_p\right\|_2 \tag{8}$$

where $\mathbf{x}_p$ and $\mathbf{x}'_p$ are the p_{th} pixel vector in $\mathbf{P}_{i,j}$ and ${}^s\mathbf{P}_{a,b}$, and $\|\mathbf{x}_p,\mathbf{x}'_p\|_2$ calculates the Euclidian distance between them. With consideration of the computational cost, the calculation

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is suggested to perform on the first q components of singular value decomposition (SVD), which maintains the relative distance of the original space under the orthogonal projection transformation [54], [55]. The determination of similarity is based on the judgment of the difference and its cause between the compared patches. Theoretically, two patches are considered to be similar, if the difference between them is caused only by noise. Otherwise, they are judged to be dissimilar. Assumed that the noise of each component of $\mathbf{X}$ obeys the Gaussian distribution, following [56], it can be transformed to a standardized normal distribution such that the calculation in (8) would yield variation of noise in the chi-square distribution under the similar condition [57], [58]. In this context, we can obtain a reference value of threshold T to evaluate $\text{dist}(\mathbf{P}_{i,j}, {}^{s}\mathbf{P}_{a,b})$ in (8). Here, we remark that, in practice, T still needs to be fine-tuned due to the uncertainty of noise distribution and subtle spectral difference. Besides, some patches should be allowed to have no similar objects under the limitation of the threshold, which can be considered that their texture features are unique in the image.

In our experiments, the above-mentioned processes do find similar patches, for instance, as shown in Fig. 1. However, the found patches usually have a slight deviation toward the one under investigation, which may lead to inaccuracies by the direct use of their information. Therefore, we propose to perform a further local search for spectra within the found nonlocal similar patches, finding the most spectrally similar pixels to the testing pixel $\mathbf{x}_{i,j}$. Therefore, after ${}^{S}\mathbf{P}$ is obtained, the set of most spectrally similar pixels ${}^{S}\mathbf{x} = \{{}^{s}\mathbf{x}_{u,v} \in {}^{s}\mathbf{P}_{a,b} | s \in [1, S]\}$ toward $\mathbf{x}_{i,j}$ can be located as follows:

$$\min_{{}^{s}\mathbf{x}_{u,v}} \text{dist}(\mathbf{x}_{i,j}, {}^{s}\mathbf{x}_{u,v}) = \sum_{b=1}^{B} \left\| x_{i,j}^{b}, {}^{s}x_{u,v}^{b} \right\|_2 \tag{9}$$

where $x_{i,j}^{b}$ and ${}^{s}x_{u,v}^{b}$ are the values of $\mathbf{x}_{i,j}$ and ${}^{s}\mathbf{x}_{u,v}$ in b_{th} band and $\|x_{i,j}^{b}, {}^{s}x_{u,v}^{b}\|_2$ calculates their Euclidian distance. As shown in Fig. 1, ${}^{s}\mathbf{x}_{u,v}$ might not be located at the center of ${}^{s}\mathbf{P}_{a,b}$ which denotes a precise extraction of information.

B. Global Spatial and Local Spectral Similarity-Based Manifold Learning GSRC

As mentioned in Section II-B, a manifold learning algorithm contributes to better characterization of hyperspectral data for SR-based classification. Therefore, in the proposed framework, the fusion and representation steps are generally performed on the NPE-processed data, with regard to the located similarity information obtained in Section III-A. In a practical application, it should be noted that not all the patches have similar structures. Some texture features may be unique in the scene and have no similar objects. Therefore, our proposed method prepares two different scenarios for different situations.

1) Local Spatial Information Only: In the first case, we consider the situation that no similar patches are found for a testing pixel. In addition to the reason mentioned earlier, it could also be due to a serious anomaly of the centered pixel in the current patch. In these cases, the algorithm should allow no similar objects to be found for certain patches but not to introduce irrelevant information. The solution is to consider only the neighborhood of the current testing pixel based on the local spatial consistency. Based on Sections II-A and II-B, it actually becomes a manifold learning-based GSR-based architecture. According to (5) and (7), the objective function can be expressed as

$$\arg\min_{\mathbf{A}} \left\| \mathbf{X}_{i,j}^{M} - \mathbf{D}^{M}\mathbf{A} \right\|_F^2 + \lambda \sum_{g \in G} \omega_g \|\mathbf{A}_g\|_2 \tag{10}$$

where $\mathbf{X}_{i,j}^{M} \in \mathbb{R}^{d \times t}$ represents an NPE-processed matrix-formed neighborhood with t pixels centered on $\mathbf{x}_{i,j}^{M}$ and $\mathbf{D}^{M}$ denotes the dictionary constructed by the samples selected from $\mathbf{X}^{M}$.

2) GSLS-^{ML}GSRC: For a testing pixel, when its similar pixels exist under the conditions of aforementioned process, we first remember their locations and then extract the corresponding vectors from $\mathbf{X}^{M}$. Following Section III-A, let ${}^{S}\mathbf{x}^{M} = \{{}^{s}\mathbf{x}_{u,v}^{M} | s \in [1, S]\}$ denote the set of the most spectrally similar pixels to $\mathbf{x}_{i,j}^{M}$ in nonlocal areas and $\mathbf{X}_{i,j}^{M}$ represent the neighboring pixels in the local area. The proposed GSLS is actually a fusion of nonlocal and local information, which refers to ${}^{S}\mathbf{x}^{M}$ and $\mathbf{X}_{i,j}^{M}$ under this case. As shown in Fig. 1, in our designed fusion framework, ${}^{S}\mathbf{x}^{M} \in \mathbb{R}^{d \times S}$ and $\mathbf{X}_{i,j}^{M}$ are combined together with their rows aligned such that the fused data can be represented as $\mathbf{X}_{f}^{M} \in \mathbb{R}^{d \times (t+S)}$. The proposed GSLS-^{ML}GSRC is in fact the group structure-based representation based on GSLS, of which the objective function can be expressed as follows:

$$\arg\min_{\mathbf{A}_f} \left\| \mathbf{X}_{f}^{M} - \mathbf{D}^{M}\mathbf{A}_f \right\|_F^2 + \lambda \sum_{g \in G} \omega_g \|\mathbf{A}_{fg}\|_2 \tag{11}$$

where $\mathbf{A}_f \in \mathbb{R}^{n \times (t+s)}$ is the coefficient matrix, including $\mathbf{A}_{fg}$ defines the group-form weight matrix. The class label of the testing pixel $\mathbf{x}_{i,j}$ can finally be determined by the class that provides the minimum residual error as follows:

$$\text{class}(\mathbf{x}_{i,j}) = \arg\min_{k} \left\| \mathbf{X}_{f}^{M} - \mathbf{D}^{M}\delta_k(\mathbf{A}_f) \right\|_F^2. \tag{12}$$

The pseudocode of the proposed GSLS-^{ML}GSRC approach is summarized in Algorithm 1.

IV. Experimental Results

In this section, the proposed GSLS-^{ML}GSRC is evaluated using three real hyperspectral data sets, of which the details are introduced in Section IV-A. The related parameter settings are reported in Section IV-B. For comparative purposes, typical SR-based models, namely SRC, JSRC, and GSRC, are first considered in our experiments. Meanwhile, we also perform SVM and SVM-based Markov random field (SVM-MRF) as comparisons to evaluate the proposed method [59]. Besides, we also develop NL-means-based SRC (NM-SRC), NLSS-based GSRC (NLSS-GSRC), and GSLS-based GSRC (GSLS-GSRC) for more comprehensive comparisons. Each value of overall accuracy (OA) and class-dependent accuracy (CA) are obtained for the evaluation of the different tested methods. The corresponding analysis and discussion of the classification results are reported in Section IV-C.

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Algorithm 1 GSLS-^{ML}GSRC

Input: $\mathbf{X}$, $\mathbf{x}_{i,j}$, d, t, υ, w, T.

Step 1-a: Manifold learning projection via NPE towards $\mathbf{x}_{i,j} \in \mathbf{X}$ to obtain the d-dimensional data $\mathbf{x}_{i,j}^{M} \in \mathbf{X}^{M}$ according to (6) and (7).

Step 1-b: A nonlocal search of $w \times w$ pixel-sized ${}^{S}\mathbf{P}$ towards $\mathbf{P}_{i,j}$ under the limitation of T according to (8) within $\upsilon \times \upsilon$ pixel-sized search window around $\mathbf{x}_{i,j}$.

Step 2: If ${}^{S}\mathbf{P}$ contains no elements, perform ***Step 3***, otherwise, perform ***Step 4***.

Step 3: Group structure-based representation towards t neighboring pixels $\mathbf{X}_{i,j}^{M}$ centered on $\mathbf{x}_{i,j}^{M}$ according to (10), and then jump to ***Step 7***.

Step 4: A local search of ${}^{S}\mathbf{x}$ towards $\mathbf{x}_{i,j}$ within ${}^{S}\mathbf{P}$ according to (9).

Step 5: Fusion of ${}^{S}\mathbf{x}^{M}$ and $\mathbf{X}_{i,j}^{M}$ to obtain $\mathbf{X}_{f}^{M}$ according to the locations of ${}^{S}\mathbf{x}$ and $\mathbf{X}_{i,j}$.

Step 6: Group structure-based representation towards $\mathbf{X}_{f}^{M}$ according to (11).

Step 7: Evaluation of the class-dependent minimum total residual error according to (4) and (12).

Output: Class label of $\mathbf{x}_{i,j}$.

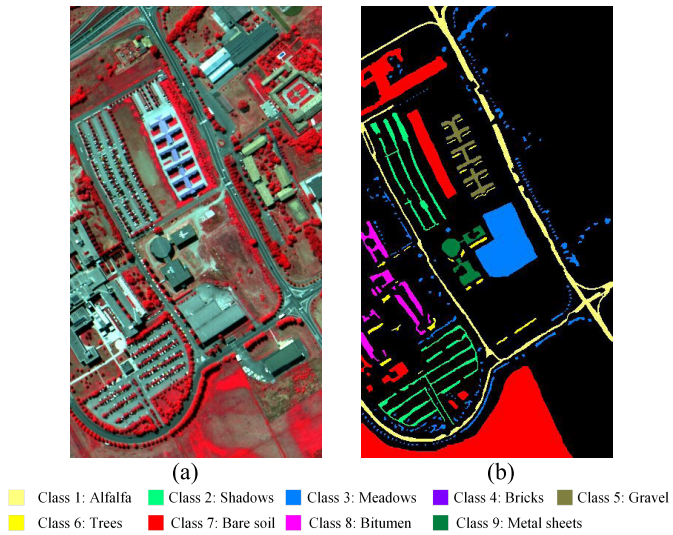


Fig. 3. ROSIS University of Pavia scene. (a) False-color composite image. (b) Reference map.

A. Data Sets' Description

1) University of Pavia Data Set: The first data set used in our experiment is the University of Pavia scene collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over an urban area surrounding the University of Pavia, Italy. The image consists of 610×340 pixels, with a spatial resolution of 1.3 m/pixel. The image contains 103 spectral bands, with the spectral coverage ranging from 0.43 to 0.86 μm, after 12 bands are removed due to the high noise. The reference of this image includes nine mutually exclusive classes, with a total of 42 776 labeled samples. The false-color composite of this image in Fig. 3(a) and (b) shows the corresponding reference data.

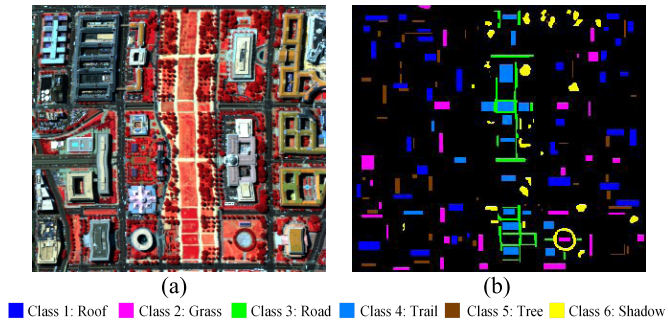


Fig. 4. HYDICE Washington DC Mall scene. (a) False-color composite image. (b) Reference map.

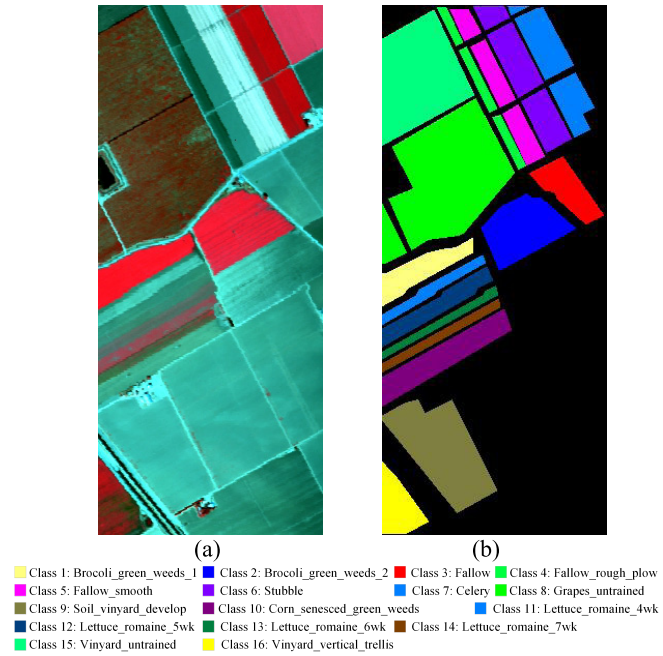


Fig. 5. AVIRIS Salinas scene. (a) False-color composite image. (b) Reference map.

2) Washington DC Mall Data Set: The second data set that we used is the Washington DC Mall scene acquired by the Hyperspectral Digital Image Collection Experiment (HYDICE) sensor over the Washington DC, USA. The image totally consists of 280×307 pixels, with a spatial resolution of 3 m/pixel. The original image contains 210 spectral bands, with the spectral coverage ranging from 0.4 to 2.5 μm. In practice, 191 bands are preserved, while the spectral bands ranging from 0.9 to 1.4 μm are removed due to the atmospheric absorption. The reference of this image includes six classes of interests, with a total of 10 190 labeled samples. Fig. 4(a) shows the false-color composite of this image, and the corresponding reference data *are* provided in Fig. 4(b).

3) Salinas Data Set: The last data set that we used is the Salinas data set captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley, CA, USA. The image totally consists of 512×217 pixels, with a geometric resolution of 3.7 m/pixel. The image contains 224 spectral bands, with the spectral coverage ranging from 0.2 to 2.4 μm, after 20 bands are removed due to the water absorption. The reference data includes

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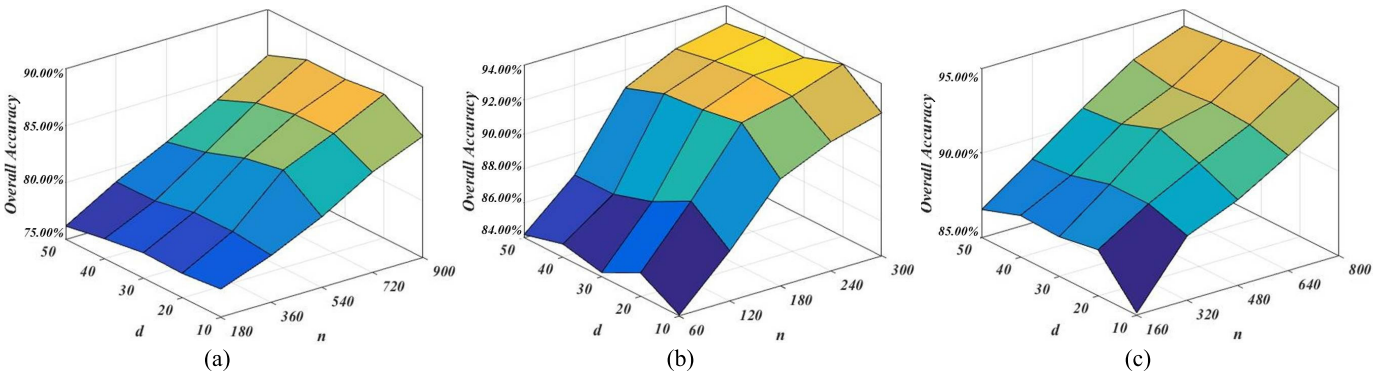


Fig. 6. Dimensionality d in NPE versus the total number n of labeled samples selected for GSR-based representation using (a) University of Pavia data set, (b) Washington DC Mall data set, and (c) Salinas data set.

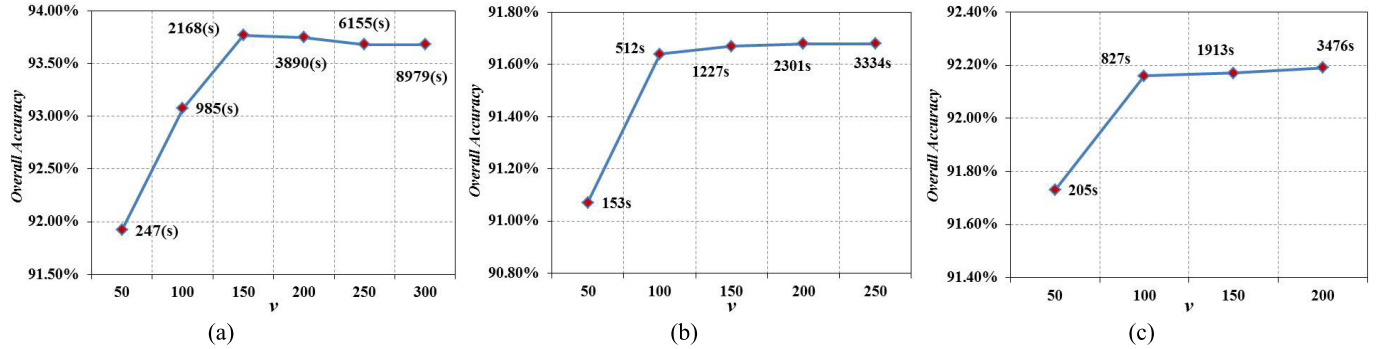


Fig. 7. Searching window size υ and the computation time (in seconds) in GSLS versus the OA results using (a) University of Pavia data set, (b) Washington DC Mall data set, and (c) Salinas data set.

16 ground-truth classes, with a total of 54 129 labeled samples. Fig. 5(a) and (b) shows the false-color composite image and the corresponding reference data, respectively.

B. Parameter Setting

1) Dimensionality in NPE: The manifold learning dimensionality d in NPE for the proposed framework is briefly investigated. Fig. 6(a)–(c) shows the OA results as a function of d and the total number of labeled samples n selected for the GSR-based classification with different data sets. According to Fig. 6, d is basically set to the value that is generally optimal or suboptimal for most cases of each data set. Therefore, d is set to 20, 20, and 30 for the University of Pavia scene, the Washington DC Mall scene, and the Salinas scene, respectively.

2) Searching Window Size: The relationships among the OA results, searching window size υ, and computational cost of different data sets are shown in Fig. 7(a)–(c). As shown in Fig. 7, when υ is extended to a certain size, OA tends to be stable, but the computation time has a relatively obvious increase. Therefore, according to Fig. 7, with a consideration of the balance between the OA results and the computational cost, υ is finally set to 150, 100, and 100 for the University of Pavia scene, the Washington DC Mall scene, and the Salinas scene, respectively.

3) Measurement of Similarity: As mentioned in Section III-A, it is suggested to first apply an SVD to preserve the first q components with consideration of the computational cost, which has been proven to be able to maintain the accuracy of the measurement of similarity. As shown in Fig. 8, in our experiment, q is selected as 4, 4, and 3 with the most concentrated power of signals for the University of Pavia scene, the Washington DC Mall scene, and the Salinas scene, respectively. After q is settled, the degree of freedom of the variation of noise can be determined according to the size w of the current assessed patch such that the threshold T can be further obtained by lookup table of the *chi*-square distribution. However, this just provides a reference value, which still needs to be fine-tuned in practice. Fig. 9 briefly shows the relationship between w and the best OA results in their cases. As shown in Fig. 9, when w reaches a certain level, the similarity information at this level is relatively reduced such that it leads to a decrease in the OA results. Therefore, w is finally selected as, 9, 7, and 9 for the University of Pavia scene, the Washington DC Mall scene, and the Salinas scene, respectively.

4) Group Structure-Based Representation: As mentioned in Section II-A, the basic assumption of GSRC is that the neighboring pixels have a high probability to belong to the same class. In this context, GSRC needs to consider a small neighborhood around the testing pixel in order to satisfy the homogeneity assumption. Therefore, we generally consider a 3×3 pixel-sized, i.e., $t = 9$, testing neighborhood in GSRC for all three data sets. Following [27] and [29], the regularization parameter λ of the representation-based methods is basically set to the values ranging from 10^{-3} to 10^{-1}. In the experiments of this article, all the tested methods are performed with

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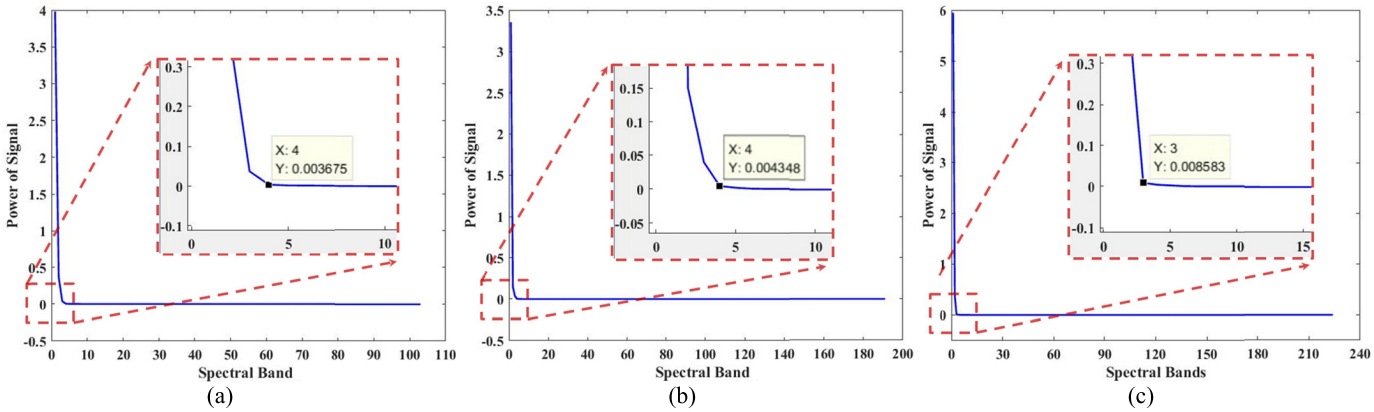


Fig. 8. Spectral band versus the power of signal in SVD using (a) University of Pavia data set, (b) Washington DC Mall data set, and (c) Salinas data set.

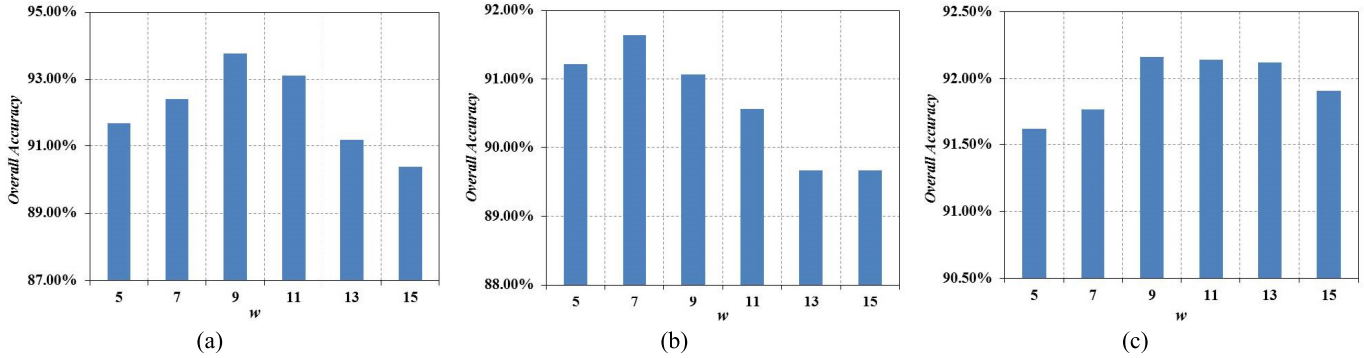


Fig. 9. Patch size w in GSLS versus the OA results using (a) University of Pavia data set, (b) Washington DC Mall data set, and (c) Salinas data set.

TABLE I

OVERALL AND CLASS-SPECIFIC ACCURACIES (IN PERCENT) OBTAINED BY THE DIFFERENT TESTED METHODS FOR THE ROSIS UNIVERSITY OF PAVIA SCENE. IN ALL CASES, 720 LABELED SAMPLES IN TOTAL (80 PER CLASS) WERE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Class	SVM	SRC	SVM-MRF	JSRC	GSRC	NM-SRC	NLSS-GSRC	GSLS-GSRC	GSLS-^{ML}GSRC
1	71.50%	72.67%	**88.65%**	62.14%	68.66%	69.91%	64.53%	68.45%	72.04%
2	70.23%	77.16%	79.49%	85.69%	87.36%	82.94%	93.82%	99.54%	**99.69%**
3	75.92%	75.18%	86.18%	83.71%	89.14%	94.09%	94.71%	97.95%	**98.33%**
4	89.82%	91.06%	92.14%	94.16%	**95.99%**	90.27%	93.83%	91.48%	92.62%
5	99.26%	93.75%	99.64%	**100.00%**	**100.00%**	99.93%	**100.00%**	**100.00%**	**100.00%**
6	78.18%	78.03%	94.17%	71.96%	82.10%	84.65%	86.80%	94.41%	**95.74%**
7	90.75%	85.04%	95.21%	96.17%	99.25%	97.89%	**99.92%**	99.62%	99.62%
8	76.22%	68.77%	91.87%	80.72%	82.40%	85.06%	90.41%	89.38%	**90.22%**
9	99.83%	89.02%	**99.88%**	97.89%	97.36%	97.36%	92.71%	95.99%	93.45%
OA	76.30%	77.77%	86.49%	82.24%	85.11%	83.70%	88.56%	92.53%	**93.42%**

respect to the same number of randomly selected labeled samples per class, which means that ω_g in GSRC equals to one.

C. Experimental Results and Analysis

1) University of Pavia Scene: In the first test with the ROSIS University of Pavia scene, we randomly select 80 labeled samples per class with a total of 720 labeled samples (which constitutes approximately 1.68% of the available labeled samples) for the construction of the dictionary in the SR-based methods and model training in the compared SVM-based methods, where the remaining samples are used for validation. Table I reports the OA and CA results obtained by different tested methods, and their corresponding classification maps are shown in Fig. 10. Based on these results, several preliminary conclusions can be drawn as follows.

1) Compared with SVM, SRC provides an evenly matched classification result, which first proves the validity of the representation-based frameworks. In addition, SVM-MRF and JSRC bring better classification performances toward SVM and SRC, which further confirms the effectiveness of combining the local spatial information. Moreover, GSRC outperforms JSRC, which demonstrates the improvement of the group structure-based priors.

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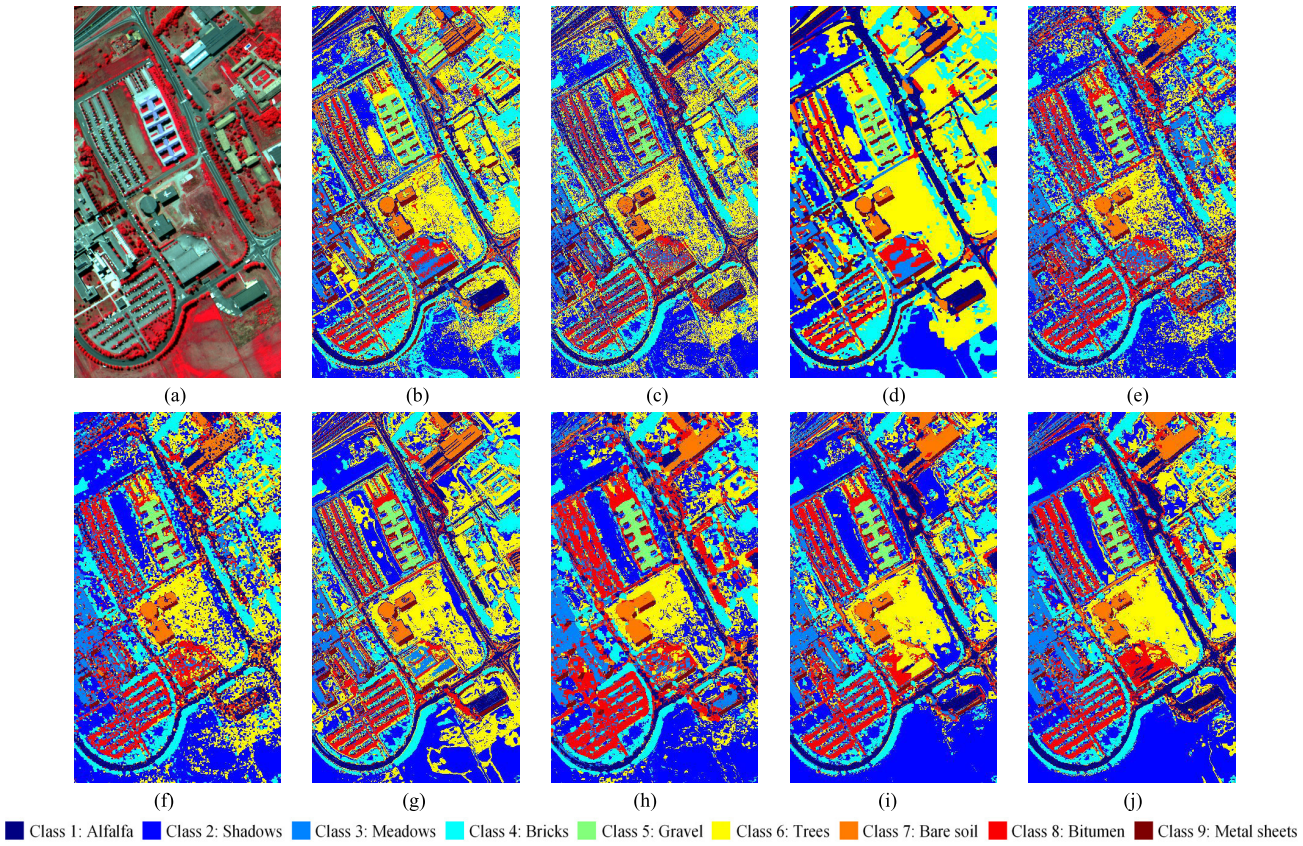


Fig. 10. Classification maps obtained by the different tested methods for the ROSIS University of Pavia data set. In all cases, 720 labeled samples in total (80 per class) were used. The overall classification accuracies are given in the parentheses. (a) False-color image. (b) SVM (76.30%). (c) SRC (77.77%). (d) SVM-MRF (86.49%). (e) JSRC (82.24%). (f) GSRC (85.11%). (g) NM-SRC (83.70%). (h) NLSS-GSRC (88.56%). (i) GSLS-GSRC (92.53%). (j) GSLS-^{ML}GSRC (93.42%).

2) Compared with SRC, NM-SRC brings an increase in the classification accuracy, which benefits from the nonlocal spatial information. In addition, NLSS-GSRC provides better classification result toward GSRC and NM-SRC, which demonstrates that the integration of nonlocal spatial information and local spatial information is effective for the classification tasks.
3) Compared with NLSS-GSRC, GSLS-GSRC yields better classification result, which indicates the improvement of exploration of local spectral similarity through nonlocal spatial similarity and the effectiveness of the designed fusion pattern for cooperative representation of the global prior information. On the other hand, the OA result of GSLS-^{ML}GSRC exceeds GSLS-GSRC, which verifies the significance of manifold learning characterization for the representation-based models.

In general, GSLS-^{ML}GSRC obtains the best classification result in comparison with the other related tested methods, with an OA of 93.42%, which is 5.77% and 2.72% higher than GSRC and NLSS-GSRC. Besides, GSLS-^{ML}GSRC brings a better expression of details within the classification map, which proves the effectiveness of the proposed method.

In the second test with the University of Pavia scene, the tested methods are evaluated by a different number of labeled samples (from 20 to 100 samples per class). As shown in Table II, we can consolidate our previous conclusions. First, both local and nonlocal spatial information bring improvements to the classification tasks, which can be seen from the results obtained by JSRC, GSRC, and NM-SRC in comparison with their original counterpart, i.e., SRC. Second, the integration of local and nonlocal spatial information contributes to more comprehensive discrimination, which can be observed from the classification performances provided by NLSS-GSRC. The exploration of local spectral similarity through nonlocal spatial similarity benefits the extraction of information and consideration of global distribution, which can be proved from the increase in results obtained by GSLS-GSRC. Finally, the designed fusion pattern and manifold learning techniques promote better characterization and representation, which can be verified from the improvement brought by GSLS-^{ML}GSRC. In all cases, the proposed method obtained the best results in comparison with other tested local or nonlocal similarity-based methods, which demonstrates its efficiency and stability.

2) Washington DC Mall Scene and Salinas Scene: In the first tests with the Washington DC Mall scene and the Salinas scene, we randomly select 10 and 20 labeled samples per class with a total of 60 and 320 samples for training and dictionary construction (which both represent approximately 0.59% of the available labeled samples), while the remaining labeled samples are used for validation. Tables III and IV

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TABLE II

OVERALL CLASSIFICATION ACCURACIES (IN PERCENT) OBTAINED BY THE DIFFERENT TESTED METHODS WITH A DIFFERENT NUMBER OF LABELED SAMPLES FOR THE ROSIS UNIVERSITY OF PAVIA SCENE. BOTH THE TOTAL NUMBER OF SAMPLES AND THE NUMBER OF SAMPLES PER CLASS ARE GIVEN (IN THE PARENTHESES)

Samples	SVM	SRC	SVM-MRF	JSRC	GSRC	NM-SRC	NLSS-GSRC	GSLS-GSRC	GSLS-^{ML}GSRC
180 (20)	65.75%	70.11%	75.67%	77.16%	79.97%	78.96%	82.14%	83.67%	85.04%
270 (30)	67.59%	72.74%	79.92%	77.53%	80.89%	79.79%	84.30%	85.90%	87.43%
360 (40)	72.76%	73.34%	80.89%	78.67%	81.88%	80.75%	85.49%	87.93%	89.28%
450 (50)	73.65%	74.58%	81.97%	79.33%	82.08%	81.31%	85.67%	88.61%	90.44%
540 (60)	74.73%	75.58%	82.33%	80.54%	83.71%	82.23%	86.17%	90.27%	91.46%
630 (70)	75.08%	76.37%	84.77%	81.27%	84.54%	83.08%	87.66%	91.18%	92.33%
720 (80)	76.30%	77.77%	86.49%	82.24%	85.11%	83.70%	88.56%	92.53%	93.42%
810 (90)	78.93%	78.22%	86.94%	83.16%	86.48%	84.97%	89.04%	92.93%	94.65%
900 (100)	80.27%	79.63%	87.51%	85.04%	87.65%	85.99%	90.70%	93.27%	95.43%

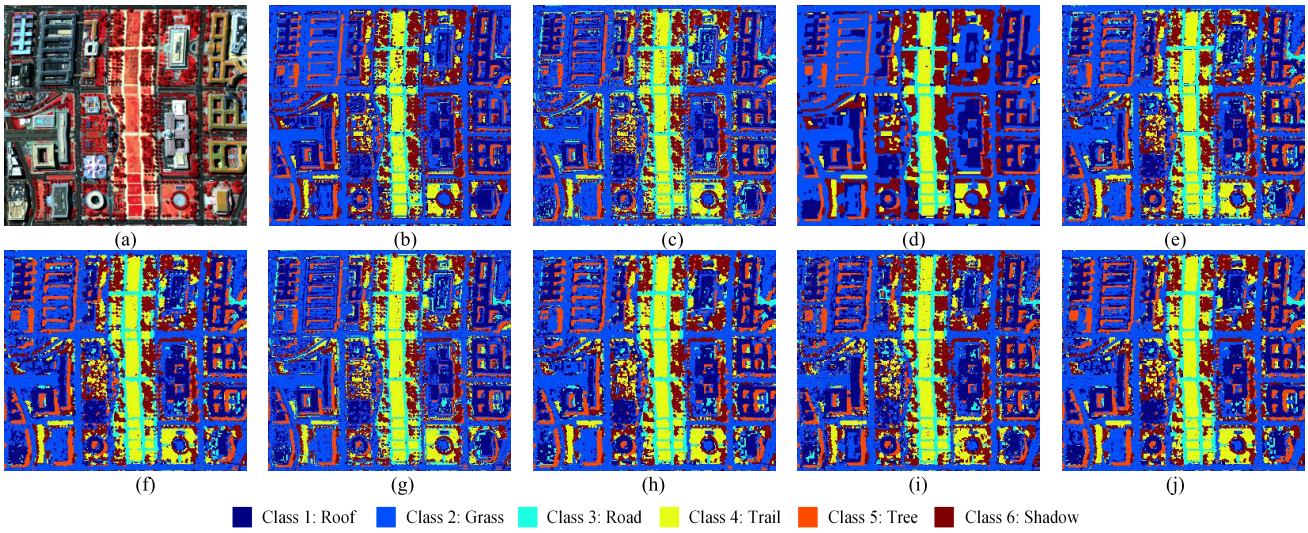


Fig. 11. Classification maps obtained by the different tested methods for the HYDICE Washington DC Mall data set. In all cases, 60 labeled samples in total (10 per class) were used. The overall classification accuracies are given in the parentheses. (a) False-color image. (b) SVM (85.31%). (c) SRC (86.40%). (d) SVM-MRF (89.30%). (e) JSRC (87.72%). (f) GSRC (89.31%). (g) NM-SRC (88.07%). (h) NLSS-GSRC (89.74%). (i) GSLS-GSRC (90.86%). (j) GSLS-^{ML}GSRC (91.64%).

TABLE III

OVERALL AND CLASS-SPECIFIC ACCURACIES (IN PERCENT) OBTAINED BY THE DIFFERENT TESTED METHODS FOR THE HYDICE WASHINGTON DC MALL SCENE. IN ALL CASES, 60 LABELED SAMPLES IN TOTAL (10 PER CLASS) WERE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Class	SVM	SRC	SVM-MRF	JSRC	GSRC	NM-SRC	NLSS-GSRC	GSLS-GSRC	GSLS-^{ML}GSRC
1	74.80%	72.33%	81.22%	73.01%	77.33%	72.33%	75.82%	82.58%	**82.03%**
2	83.43%	89.33%	90.53%	93.02%	93.90%	92.91%	92.85%	89.94%	**95.11%**
3	87.28%	94.07%	89.13%	95.97%	94.15%	95.73%	**99.21%**	95.65%	98.42%
4	91.87%	93.85%	95.95%	93.91%	93.30%	96.42%	**96.87%**	95.08%	94.36%
5	**98.10%**	88.75%	95.58%	90.09%	94.38%	90.80%	91.79%	93.93%	94.82%
6	91.31%	94.22%	94.77%	94.85%	95.32%	95.39%	95.94%	**97.74%**	95.32%
OA	85.51%	86.40%	89.30%	87.72%	89.31%	88.07%	89.74%	90.86%	**91.64%**

report the OA and CA results of different tested methods, where Figs. 11 and 12 show the corresponding classification maps for two data sets, respectively. As shown in Tables III and IV, similar conclusions can be conducted with the University of Pavia's experiment. On one hand, GSRC, JSRC, and SVM-MRF achieve an increase in classification results relative to SRC and SVM by integrating the local spatial consistency. On the other hand, GSLS-GSRC, NLSS-GSRC, and NM-SRC also bring improvements to their original counterparts with consideration of nonlocal spatial similarity. Most importantly, the proposed GSLS-^{ML}GSRC obtains the best classification result with OA results of 91.64% and 93.79%, which is 5.24% and 11.94% higher than SRC for two data sets, respectively. In addition, it also provides better performances for individual classes, which can be seen from the classification maps shown in Figs. 11 and 12.

The second tests with the Washington DC Mall scene and the Salinas scene evaluate the proposed GSLS-^{ML}GSRC method with a varying size of labeled samples (from 10 to 50 samples per class). Figs. 13 and 14 show the OA results

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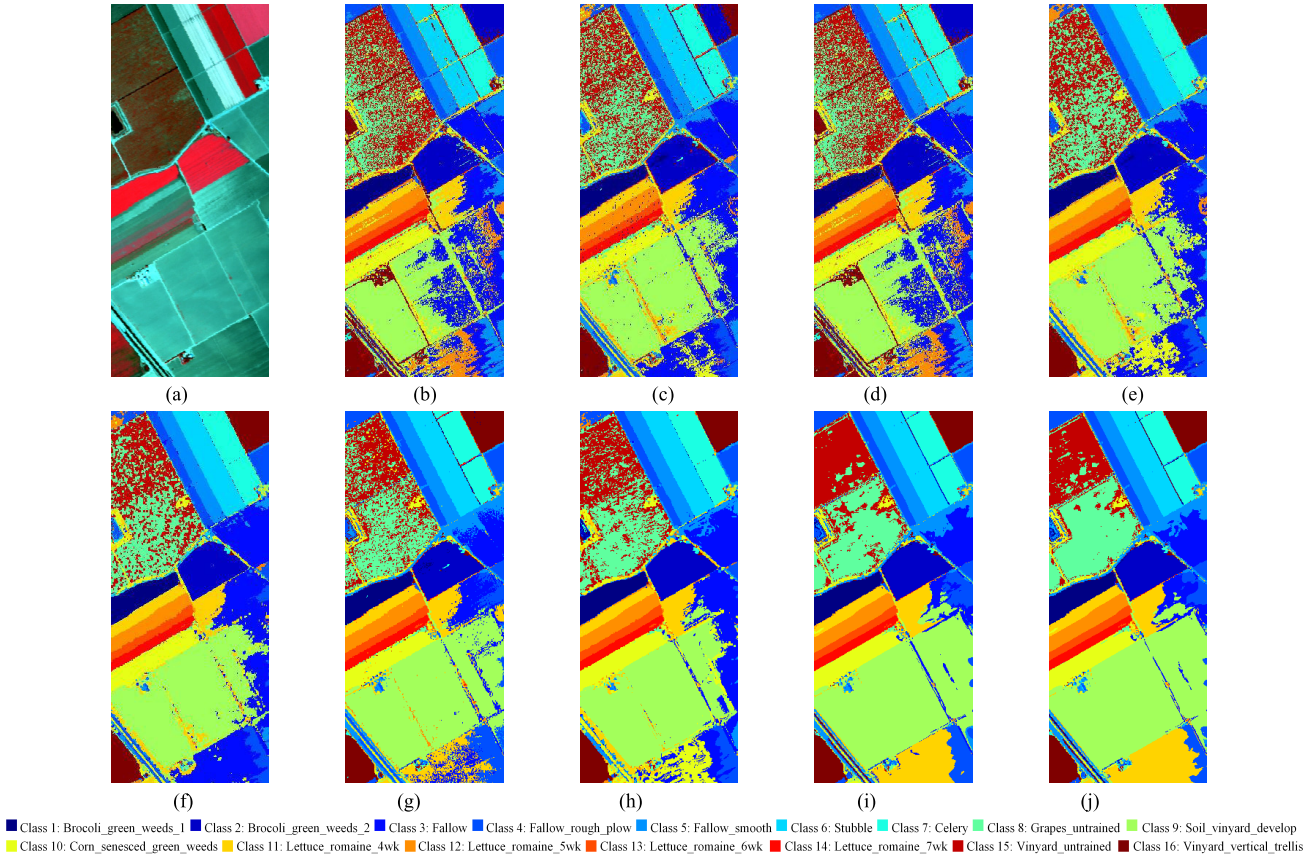


Fig. 12. Classification maps obtained by the different tested methods for the AVIRIS Salinas data set. In all cases, 320 labeled samples in total (20 per class) were used. The overall classification accuracies are given in the parentheses. (a) False-color image. (b) SVM (81.63%). (c) SRC (81.85%). (d) SVM-MRF (85.27%). (e) JSRC (84.70%). (f) GSRC (87.56%). (g) NM-SRC (86.87%). (h) NLSS-GSRC (89.39%). (i) GSLS-GSRC (92.18%). (j) GSLS-^{ML}GSRC (93.79%).

TABLE IV

OVERALL AND CLASS-SPECIFIC ACCURACIES (IN PERCENT) OBTAINED BY THE DIFFERENT TESTED METHODS FOR THE AVIRIS SALINAS SCENE. IN ALL CASES, 320 LABELED SAMPLES IN TOTAL (20 PER CLASS) WERE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Class	SVM	SRC	SVM-MRF	JSRC	GSRC	NM-SRC	NLSS-GSRC	GSLS-GSRC	GSLS-^{ML}GSRC
1	94.49%	99.35%	96.45%	99.85%	**100.00%**	99.60%	**100.00%**	**100.00%**	**100.00%**
2	96.18%	91.47%	97.09%	97.05%	99.17%	96.27%	99.14%	99.30%	**99.33%**
3	90.20%	84.36%	93.90%	83.15%	**97.42%**	73.48%	80.16%	70.24%	78.24%
4	95.38%	97.20%	94.77%	99.21%	99.21%	99.14%	99.35%	**99.64%**	**99.64%**
5	94.87%	96.38%	95.10%	99.03%	98.62%	99.25%	99.10%	99.51%	**99.74%**
6	95.85%	98.81%	92.88%	**100.00%**	**100.00%**	99.87%	**100.00%**	99.87%	99.92%
7	94.56%	98.35%	95.28%	99.80%	99.83%	99.44%	**99.94%**	99.64%	99.75%
8	55.18%	55.99%	65.41%	59.12%	63.85%	67.67%	70.98%	77.88%	**86.82%**
9	91.03%	96.03%	95.87%	98.90%	99.31%	99.21%	99.87%	99.95%	**99.98%**
10	76.16%	80.63%	81.78%	88.13%	**94.69%**	88.96%	92.13%	90.51%	90.42%
11	87.79%	95.79%	90.38%	99.53%	**99.91%**	98.50%	99.53%	95.60%	95.88%
12	96.81%	98.29%	98.98%	**100.00%**	99.84%	99.84%	**100.00%**	**100.00%**	**100.00%**
13	94.52%	93.78%	95.56%	99.56%	98.69%	98.03%	**99.89%**	99.24%	99.56%
14	88.20%	90.00%	91.02%	95.33%	**96.73%**	94.49%	96.45%	96.26%	95.42%
15	55.82%	61.60%	51.81%	63.68%	69.37%	69.74%	76.91%	**90.97%**	86.81%
16	91.07%	96.35%	95.22%	98.51%	98.51%	98.45%	**98.62%**	98.67%	98.56%
OA	81.63%	81.85%	85.27%	84.70%	87.56%	86.87%	89.39%	92.18%	**93.79%**

provided by different tested methods, as a function of the number of labeled samples used for training and dictionary construction for two data sets, respectively. As shown in Figs. 13 and 14, the superiority of the proposed method is confirmed by the best classification results obtained by GSLS-^{ML}GSRC in all cases.

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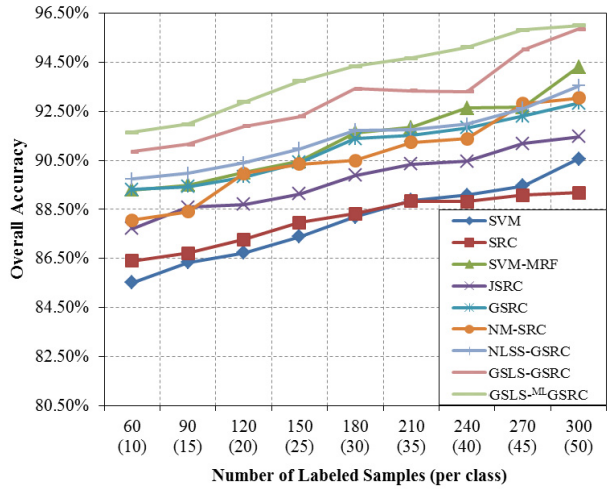


Fig. 13. Overall classification results obtained by the different tested methods as a function of the number of labeled samples (per class) for the HYDICE Washington DC Mall scene.

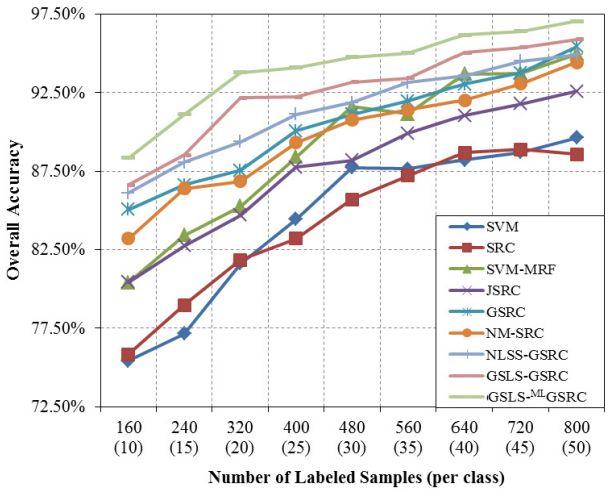


Fig. 14. Overall classification results obtained by the different tested methods as a function of the number of labeled samples (per class) for the AVIRIS Salinas scene.

V. Conclusion

In this article, a novel GSLS-based manifold learning GSR classifier, called as GSLS-^{ML}GSRC, is proposed for hyperspectral imagery. The main contributions of the proposed method first depend on the exploration of local spectral similarity through nonlocal spatial similarity, followed by a designed pattern to fuse the nonlocal and local spatial information, finally located at the optimization of searching mode and characterization by manifold learning algorithm for better representation. Experimental results on three real hyperspectral data sets demonstrate that the proposed GSLS-^{ML}GSRC outperforms other related nonlocal or local similarity-based methods for classification.

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Haoyang Yu (S'16–M'19) received the B.S. degree in information and computing science from Northeastern University, Shenyang, China, in 2013, and the Ph.D. degree in cartography and geographic information system from the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, China, in 2019.

He is currently a Xing Hai Associate Professor with the Center of Hyperspectral Imaging in Remote Sensing (CHIRS), Information Science and Technology College, Dalian Maritime University, Dalian, China. His research interests include models and algorithms for hyperspectral image processing, analysis, and applications.

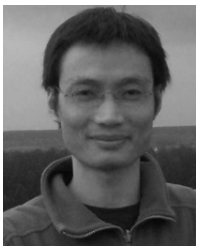


Lianru Gao (M'12–SM'18) received the B.S. degree in civil engineering from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He was a Visiting Scholar with the University of Extremadura, Cáceres, Spain, in 2014, and Mississippi State University (MSU), Starkville, MS, USA, in 2016. He is currently a Professor with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, CAS. In last ten years, he was a PI of ten scientific research projects at national and ministerial levels, including the projects by the National Natural Science Foundation of China from 2010 to 2012, from 2016 to 2019, and from 2018 to 2020 and the Key Research Program of the CAS from 2013 to 2015. He was supported by the China National Science Fund for Excellent Young Scholars in 2017. He has authored or coauthored more than 130 peer-reviewed articles, and there are 60 journal articles included by the Science Citation Index (SCI). He has coauthored an academic book *Hyperspectral Image Classification and Target Detection*. He holds 22 National Invention Patents and four Software Copyright Registrations in China. His research interests include models and algorithms for hyperspectral image processing, analysis, and applications.

Dr. Gao was a recipient of the Outstanding Science and Technology Achievement Prize of the CAS in 2016, the Second Prize of The State Scientific and Technological Progress Award in 2018, the recognition of the Best Reviewers of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2015, and the Best Reviewers of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2017.

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Wenzhi Liao (S'10–M'14–SM'16) received the B.Sc. degree in mathematics from Hainan Normal University, Haikou, China, in 2006, the Ph.D. degree in engineering from the South China University of Technology, Guangzhou, China, in 2012, and the Ph.D. degree in computer science engineering from Ghent University, Ghent, Belgium, in 2012.

From 2012 to 2019, he has been holding a post-doctoral position at Ghent University and then a Post-Doctoral Research Fellow with the Research Foundation Flanders (FWO-Vlaanderen), Brussels, Belgium. Since 2019, he has been a Lecturer with the University of Strathclyde, Glasgow, U.K. His research interests include pattern recognition, remote sensing, image processing, mathematical morphology, multitask feature learning, multisensor data fusion, and hyperspectral image restoration.

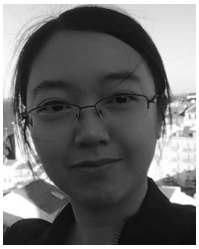
Dr. Liao was a recipient of the Best Paper Challenge Award from the 2013 IEEE GRSS Data Fusion Contest and the 2014 IEEE GRSS Data Fusion Contest. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *IET Image Processing*.



Bing Zhang (M'11–SM'12) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

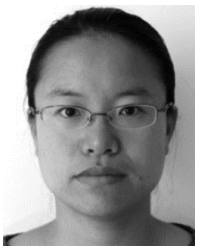
He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS, where he has been leading key scientific projects in the area of hyperspectral remote sensing for more than 20 years. He has developed five software systems in the image processing and applications. He has authored more than 300 publications, including more than 170 journal articles. He has edited six books/contributed book chapters on hyperspectral image processing and subsequent applications. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang has been serving as a Technical Committee Member for the IEEE Workshop on Hyperspectral Image and Signal Processing since 2011 and the President of the Hyperspectral Remote Sensing Committee of the China National Committee of International Society for Digital Earth since 2012. He is a member of the Student Paper Competition Committee, IGARSS 2015, 2016, and 2017. His creative achievements were rewarded ten important prizes from the Chinese government and special government allowances of the Chinese State Council. He was a recipient of the National Science Foundation for Distinguished Young Scholars of China in 2013, the 2016 Outstanding Science and Technology Achievement Prize of the Chinese Academy of Sciences, and the highest level of awards for the CAS scholars. He serves as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Lina Zhuang (S'15) received the bachelor's degrees in geographic information system and in economics from South China Normal University, Guangzhou, China, in 2012, the M.S. degree in cartography and geography information system from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2015, and the Ph.D. degree in electrical and computer engineering from the Instituto Superior Tecnico, Universidade de Lisboa, Lisbon, Portugal, in 2018.

Since 2015, she has been a Marie Curie Early Stage Researcher of the Sparse Representations and Compressed Sensing Training Network (SpaRTaN number 607290) with the Instituto de Telecomunicacoes. SpaRTaN Initial Training Networks (ITN) is funded under the European Unions Seventh Framework Programme (FP7-PEOPLE-2013-ITN) call and is part of the Marie Curie Actions–ITN funding scheme. She is currently a Research Assistant Professor with Hong Kong Baptist University, Hong Kong. Her research interests include hyperspectral image restoration, super-resolution, and compressive sensing.



Meiping Song received the Ph.D. degree from the College of Computer Science and Technology, Harbin Engineering University, Harbin, China, in 2006.

From 2013 to 2014, she was a Visiting Associate Research Scholar at the University of Maryland at Baltimore, Baltimore, MD, USA. She is currently an Associate Professor with the Information Science and Technology College, Dalian Maritime University, Dalian, China. Her research interests include remote sensing and hyperspectral image processing.



Jocelyn Chanussot (M'04–SM'04–F'12) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

In 1999, he was with the Geography Imagery Perception Laboratory for the Delegation Generale de l'Armement (DGA–French National Defense Department), Arcueil, France. Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He was a Visiting Scholar with Stanford University, Stanford, CA, USA, KTH, Stockholm, Sweden, and the National University of Singapore, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles, Los Angeles, CA, USA. He is conducting his research at the GIPSA-Lab. His research interests include image analysis, multicomponent image processing, nonlinear filtering, and data fusion in remote sensing.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008 and the Institut Universitaire de France from 2012 to 2017. He was the founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Co-Chair from 2005 to 2008 and the Chair from 2009 to 2011 of the GRS Data Fusion Technical Committee. He was an Associate Editor of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2005 to 2007 and *Pattern Recognition* from 2006 to 2008. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. Since 2007, he has been serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was a Guest Editor of the PROCEEDINGS OF THE IEEE in 2013 and the *IEEE Signal Processing Magazine* in 2014. Since 2018, he has also been serving as an Associate Editor for the IEEE TRANSACTIONS ON IMAGE PROCESSING.

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